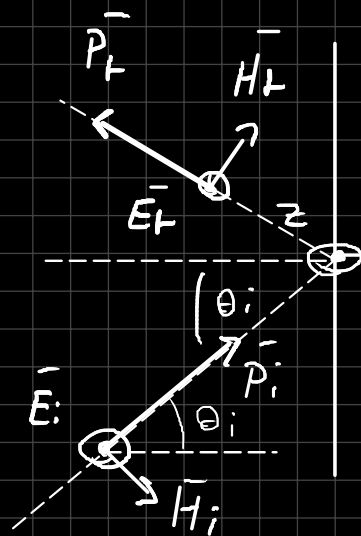


Seguimos con incidencia oblicua:

dirección



Conductor perfecto, $\vec{P}_i = |\vec{P}_i| (\sin \theta_i \hat{x} + \cos \theta_i \hat{z})$
 NO HAY CAMPOS

$$\vec{P}_r = |\vec{P}_r| (\sin \theta_i \hat{x} - \cos \theta_i \hat{z})$$

Llamamos $\vec{r} = \beta (\sin \theta_i \hat{x} + \cos \theta_i \hat{z})$

Podemos escribir $\vec{E}_i$ así:

Diel. Perf

$$\vec{E}_i = E_i e^{-j \vec{k} \cdot \vec{r}} \hat{y} \quad \text{donde } \vec{r} = x \hat{x} + y \hat{y} + z \hat{z}$$

$$\Rightarrow \vec{E}_i = E_i e^{-j \beta (x \sin \theta_i + z \cos \theta_i)} \hat{y}$$

$\vec{H}_i$? $\rightarrow$ Opción 1) $\vec{H}_i = - \frac{\nabla \times \vec{E}_i}{j \omega \mu}$ Opción 2) sacarlo del dibujo

$$\begin{aligned}
 \bar{H}_i &= H_i e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)} \cdot (-\cos \theta_i \hat{x} + \sin \theta_i \hat{z}) = \overset{H_i = \frac{E_i}{Z_1}}{} \\
 &= \frac{E_i}{Z_1} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)} (-\cos \theta_i \hat{x} + \sin \theta_i \hat{z})
 \end{aligned}$$

Para $\bar{E}_t$ haremos lo mismo:

$$\bar{E}_t = E_t e^{-j\beta_1(x \sin \theta_i - z \cos \theta_i)} \hat{y}$$

Idem $\bar{H}_t$:

$$\bar{H}_t = \frac{E_i}{Z_1} \cdot e^{-j\beta_1(x \sin \theta_i - z \cos \theta_i)} \cdot (\cos \theta_i \hat{x} + \sin \theta_i \hat{y})$$

Para la frontera $z=0$ podemos aplicar la condición de contorno:

$$E_{\text{tang}} = E_i + E_r = 0 \Rightarrow E_r = -E_i \rightarrow \text{Se invierte la fase}$$

$$(\Gamma = -1)$$

$$\begin{aligned} E_r &= E_i \cdot e^{-j\beta_1 (x \sin \theta_i + z \cos \theta_i)} \hat{y} - E_i \cdot e^{-j\beta_1 (x \sin \theta_i - z \cos \theta_i)} \hat{y} = \\ &= E_i \cdot e^{-j\beta_1 x \sin \theta_i} \cdot \left[e^{-j\beta_1 z \cos \theta_i} - e^{j\beta_1 z \cos \theta_i} \right] \cdot \hat{y} \\ &= E_i \cdot e^{-j\beta_1 x \sin \theta_i} \cdot (-2j) \sin(\beta_1 z \cos \theta_i) \cdot \hat{y} \end{aligned}$$

Para $\vec{H}_r$ se puede hacer lo mismo, pero son muchas más cuentas

$$\begin{aligned} \vec{H}_i &= \frac{E_i}{Z_1} (-\cos \theta_i \hat{x} + \sin \theta_i \hat{z}) e^{-j\beta_1 (x \sin \theta_i + z \cos \theta_i)} \rightarrow \text{incidente} \\ &- \frac{E_i}{Z_1} (\cos \theta_i \hat{x} + \sin \theta_i \hat{z}) e^{-j\beta_1 (x \sin \theta_i + z \cos \theta_i)} \rightarrow \text{reflejado} \end{aligned}$$

se puede demostrar que:

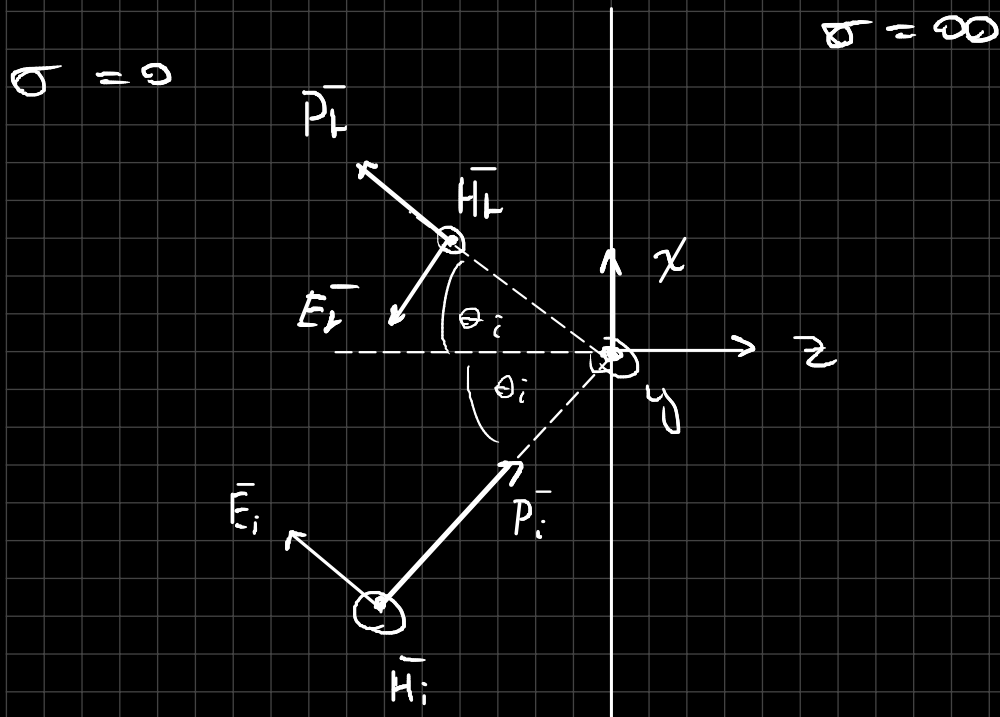
$$\bar{H}_1 = - \frac{2 E_i}{z_1} \left[\hat{x} \cos \theta_i \cos(\beta_1 z \cos \theta_i) + \hat{z} j \sin \theta_i \sin(\beta_1 z \cos \theta_i) \right]$$

también se puede demostrar que:

$$\langle \bar{P}_1 \rangle = \frac{1}{2} \operatorname{Re} \{ \bar{E}_1 \times \bar{H}_1^* \} = 2 \frac{E_i^2}{z_1} \sin^2(\beta_1 z \cos \theta_i) \cdot \sin \theta_i \cdot \hat{x}$$

La potencia apunta
en $\hat{x}$ y avanza en z .

Polarización paralela



$$\vec{E}_i = E_i e^{-j\beta_1(x\sin\theta_i + z\cos\theta_i)} \cdot (\cos\theta_i \hat{x} - \sin\theta_i \hat{z})$$

$$\vec{H}_i = H_i e^{-j\beta_1(x\sin\theta_i + z\cos\theta_i)} \hat{y}$$

$$\vec{E}_L = E_L e^{-j\beta_1 (1 \cos \theta; x - \cos \theta; z)} (-\cos \theta; \hat{x} - 1 \sin \theta; \hat{z})$$

$$\vec{H}_L = H_L e^{-j\beta_1 (1 \cos \theta; x - \cos \theta; z)} \hat{y}$$

Condição de contorno. Em $z=0$ $E_{\text{tang}} = E_{i \text{ tang}} + E_{L \text{ tang}} = 0$

Estas são as duas componentes em x

$$\Rightarrow E_i \cancel{\cos \theta} e^{-j\beta_1 x \cancel{1 \sin \theta}} - E_L \cancel{\cos \theta} e^{-j\beta_1 x \cancel{1 \sin \theta}} = 0$$

$$\Rightarrow E_L = E_i \rightarrow \Gamma = 1$$

Una vez más, sumamos los campos $\vec{E}_i$ y $\vec{E}_r$ para hallar $\vec{E}_t$ total:

$$\begin{aligned}\vec{E}_t &= E_i (\cos \theta_i \hat{x} - \sin \theta_i \hat{z}) e^{-j\beta_1 (x \sin \theta_i + z \cos \theta_i)} + \\ &+ E_i (-\cos \theta_i \hat{x} - \sin \theta_i \hat{z}) e^{-j\beta_1 (x \sin \theta_i - z \cos \theta_i)} = \\ &= E_i e^{-j\beta_1 x \sin \theta_i} [-\hat{x} \cos \theta_i \mp j \sin (\beta_1 z \cos \theta_i) - \\ &\quad - \hat{z} \sin \theta_i \mp \cos (\beta_1 z \cos \theta_i)]\end{aligned}$$

Cuanto a...

Repetimos para $\bar{H}_1$

$$\bar{H}_1 = H_i e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)} \hat{y} + H_r e^{-j\beta_1(x \sin \theta_i - z \cos \theta_i)} \hat{y} =$$

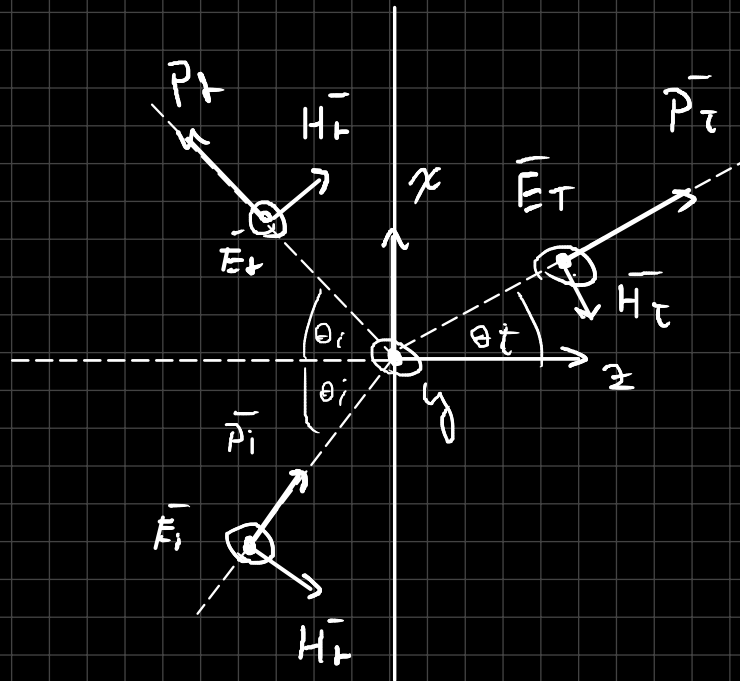
$$= 2 \frac{E_i}{Z_1} \cos(\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i} \hat{y}$$

Potencia:

$$\langle \vec{P} \rangle = \frac{1}{2} \operatorname{Re} \{ \vec{E}_1 \times \vec{H}_1^* \} = 2 \frac{E_i^2}{Z_1} \cos^2(\beta_1 z \cos \theta_i) \sin \theta_i \hat{x}$$

←
←
OTRA VEZ. ONDA
GUIADA

SIGUIENTE CASO: Dielectric vs. Dielectric.



$$\vec{E}_i = E_i e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)} \hat{y}$$

$$\vec{H}_i = \frac{E_i}{Z_1} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)} (-\cos \theta_i \hat{x} + \sin \theta_i \hat{z})$$

$$\vec{E}_r = E_r e^{-j\beta_1(x \sin \theta_r + z \cos \theta_r)} \hat{y}$$

$$\vec{H}_r = \frac{E_r}{Z_1} e^{-j\beta_1(x \sin \theta_r + z \cos \theta_r)} (\cos \theta_r \hat{x} + \sin \theta_r \hat{z})$$

$$\sigma = 0, \epsilon_1$$

$$\sigma = 0, \epsilon_2$$

Ahora además hay campos transmitidos:

$$\vec{E}_t = E_t e^{-j\beta_2(x \sin \theta_t + z \cos \theta_t)} \hat{y}$$

$$\vec{H}_t = \frac{E_t}{Z_2} e^{-j\beta_2(x \sin \theta_t + z \cos \theta_t)} (\cos \theta_t (-\hat{x}) + \sin \theta_t \hat{z})$$

condiciones de borde:

$$E_{tang1} = E_{tang2} \text{ en } z=0$$

$$\rightarrow E_i e^{-j\beta_1 x \sin \theta_i} + E_r e^{-j\beta_1 x \sin \theta_i} = E_t e^{-j\beta_2 x \sin \theta_t}$$

Campo $\vec{E}$

$$\rightarrow \frac{E_i}{Z_1} (-\cos \theta_i) e^{-j\beta_1 x \sin \theta_i} + \frac{E_r}{Z_1} \cos \theta_i e^{-j\beta_1 x \sin \theta_i} = \frac{E_t}{Z_2} (-\cos \theta_t) e^{-j\beta_2 x \sin \theta_t}$$

Campo $\vec{H}$

La ecuación del campo $\vec{E}$ la podemos escribir como:

$$(E_i + E_r) \cdot e^{-j\beta_1 x \sin \theta_i} = E_t e^{-j\beta_2 x \sin \theta_t}$$

$$\beta_1 \sin \theta_i = \beta_2 \sin \theta_t \rightarrow \text{Ley de S.D. Snell}$$

$$\left\{ \begin{array}{l} E_i + E_r = E_t \quad (1) \\ -\frac{E_i}{Z_1} \cos \theta_i + \frac{E_r}{Z_1} \cos \theta_i = -\frac{E_t}{Z_2} \cos \theta_t \quad (2) \end{array} \right.$$

$$(1) + \frac{Z_2}{\cos \theta_t} \cdot (2):$$

$$E_i + E_r - E_i \cdot \frac{Z_2}{Z_1} \frac{\cos \theta_i}{\cos \theta_t} + E_r \frac{Z_2}{Z_1} \frac{\cos \theta_i}{\cos \theta_t} = 0$$

$$E_r \left(\frac{Z_2 \cos \theta_i}{Z_1 \cos \theta_t} + 1 \right) = E_i \left(\frac{Z_2 \cos \theta_i}{Z_1 \cos \theta_t} - 1 \right)$$

$$E_r (Z_2 \cos \theta_i + Z_1 \cos \theta_t) = E_i (Z_2 \cos \theta_i - Z_1 \cos \theta_t)$$

Definimos $\Gamma_{\perp}$:

$$\Gamma_{\perp} = \frac{E_{\perp}}{E_i} = \frac{-\cos\theta_t \cdot z_1 + \cos\theta_i \cdot z_2}{\cos\theta_t z_1 + \cos\theta_i \cdot z_2}$$

Ahora buscamos la relación E_t/E_i

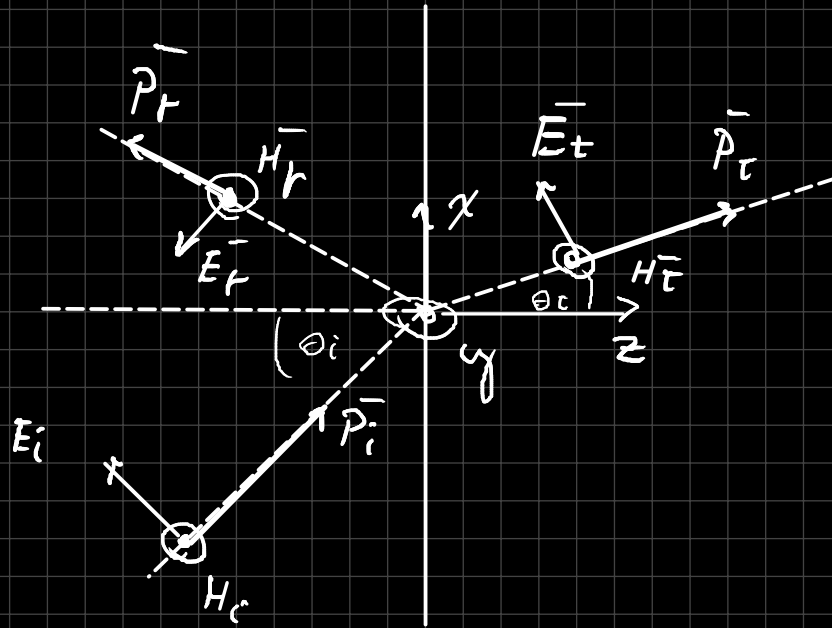
$$(1) - \frac{z_1}{\cos\theta_i} (2):$$

$$E_i + \cancel{E_r} + \cancel{E_i} - E_r = E_t + E_t \frac{z_1}{z_2} - \frac{\cos\theta_t}{\cos\theta_i}$$

$$\tau_{\perp} = \frac{E_t}{E_i} = \frac{2 \cos\theta_i z_2}{z_1 \cos\theta_t + z_2 \cos\theta_i}$$

Obs: Si $\theta_i = \theta_t = 0$, es e/ τ de
incidencia normal

Repetimos para polarización paralela



$$\sigma = 0, \epsilon_1$$

$$\sigma = 0, \epsilon_2$$

$$\vec{E}_i = E_i (\hat{x} \cos \theta_i - \hat{z} \sin \theta_i) e^{-j\beta_1 (x \sin \theta_i + z \cos \theta_i)}$$

$$\vec{H}_i = H_i \hat{y} e^{-j\beta_1 (x \sin \theta_i + z \cos \theta_i)}$$

$$\bar{E}_i = E_i (-\hat{x} \cos \theta_i - \hat{z} \sin \theta_i) e^{-j\beta_1 (x \sin \theta_i - z \cos \theta_i)}$$

$$\bar{H}_i = H_i \hat{y} e^{-j\beta_1 (x \sin \theta_i - z \cos \theta_i)}$$

$$\bar{E}_t = E_t (\hat{x} \cos \theta_t - \hat{z} \sin \theta_t) e^{-j\beta_2 (x \sin \theta_t + z \cos \theta_t)}$$

$$\bar{H}_t = H_t \hat{y} e^{-j\beta_2 (x \sin \theta_t + z \cos \theta_t)}$$

condición de borde:

$$E_i \cos \theta_i e^{-j\beta_1 x \sin \theta_i} - E_r \cos \theta_i e^{-j\beta_1 x \sin \theta_i} = E_t \cos \theta_t e^{-j\beta_2 x \sin \theta_t}$$

$$\Rightarrow \begin{cases} \beta_1 \sin \theta_i = \beta_2 \sin \theta_t \\ E_i \cos \theta_i - E_r \cos \theta_i = E_t \cos \theta_t \end{cases}$$

$$H_i e^{-j\beta_1 x \sin \theta_i} + H_r e^{-j\beta_1 x \sin \theta_i} = H_t e^{-j\beta_2 x \sin \theta_t}$$

$$\left\{ \begin{array}{l} \frac{E_i}{z_1} + \frac{E_r}{z_1} = \frac{E_t}{z_2} \quad (2) \\ E_i \cos \theta_i - E_r \cos \theta_i = E_t \cos \theta_t \quad (1) \end{array} \right. \rightarrow \text{hacer lo de siempre}$$

$$(1) - (2) \times \cos \theta_t \cdot z_2$$

$$\frac{E_r}{E_i} = \frac{\cos \theta_i \cdot z_1 - \cos \theta_t \cdot z_2}{\cos \theta_i \cdot z_1 + \cos \theta_t \cdot z_2}$$

$$\Gamma_{||} = \boxed{-} \frac{E_r}{E_i} = \frac{-z_1 \cos \theta_i + z_2 \cos \theta_t}{z_1 \cos \theta_i + z_2 \cos \theta_t}$$

→ hay q' definirlo así xq' hay inversión de base

$$\tau_{//} = \frac{2z_2 \cos \theta_i}{z_1 \cos \theta_i + z_2 \cos \theta_t}$$